

AI FITNESS & WELLNESS PLATFORM

Master Architecture & Migration Plan — Phase 0 through Phase 4

Foundation → Business MVP (incl. Integration Layer & Migration) → Coaching → AI → Advanced AI

Working reference for development — consolidated single source of truth

Status: Living document — updated as client discovery continues

1. Executive Summary

This is the single, consolidated architecture reference for the AI Fitness & Wellness Platform — covering all five build phases, the Integration Layer needed to connect external services, and the plan for migrating the first real client (Sweatfit Wellness, currently running on a system called “BPC Admin”) onto this platform.

This document merges three things that were previously separate: the original 60-section product scope, the phased architecture plan (Phase 0-4), and everything learned from reviewing the client's existing system, database schema, and third-party integrations. Where something is still unconfirmed (pending source code, API docs, or client answers), it is explicitly marked — this document does not guess at things we don't yet know.

How to read this document

- Sections 6-10 (Phase 0-4) are the core build plan — what gets built, in what order, and why.
- Section 11 (Integration Layer) explains how external services (payments, messaging, calling, access control) connect without being hardcoded into core modules.
- Section 12 (Existing System & Migration) covers the Sweatfit Wellness-specific migration — read this before touching any of their data.
- Section 16 (Open Questions) is the running list of things to confirm with the client/senior dev before development on that piece begins.

2. Guiding Principles

Four architectural decisions shape everything in this plan. These are the points most worth discussing in the review, since they affect cost, timeline, and flexibility.

2.1 The Closed-Loop Model

Assess → Plan → Train → Observe → Adapt. Phase 0-2 builds the data and workflow foundation for this loop; Phase 3-4 adds the AI that closes it automatically.

2.2 Member 360 as the Single Source of Truth

Every module — CRM, billing, assessments, programs, nutrition — writes into one unified member data profile. This is a database design principle, not a UI screen: it is what allows AI features later to reason over one consistent picture of a member instead of stitching together disconnected tables.

2.3 Modular Monolith, Not Microservices (initially)

The platform will be built as a single deployable application, internally organized into clean, independent modules (billing, scheduling, assessments, etc.). This avoids the operational overhead of microservices (separate deployments, service-to-service networking, distributed monitoring) that isn't justified at this stage. Modules are still designed with clear boundaries so specific pieces (e.g. computer vision inference in Phase 4) can be split into their own service later without a rewrite.

2.4 Multi-Tenancy & Event Logging from Day One

Every table in the system is designed from the start to isolate data per client organization (tenant), and every meaningful action (missed workout, payment, assessment completed) is logged as an event. Both are extremely

expensive to retrofit later, and event history logged early becomes training/reference data for the AI features in Phase 3-4.

3. System Architecture Overview

The platform is organized into three towers — Business Platform, Coaching Platform, and AI Platform — all reading from and writing to one Member 360 profile, sitting on Core Platform Services. New in this version: a dedicated Integration Layer underneath, connecting the platform to every external service the business actually depends on.

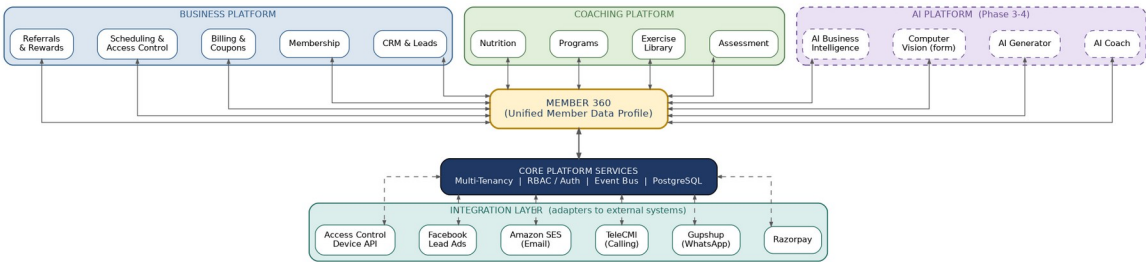


Figure 1 — High-level system architecture, including the Integration Layer added after reviewing the client's existing third-party dependencies.

One important distinction worth flagging clearly: “Access Control” (the Business Platform box) and “Computer Vision” (the AI Platform box) are NOT the same thing, even though both involve devices. Access Control is the door/gate entry hardware (QR or card scan to open a gym door) — it's a live operational need from Phase 1 onward because the client's real gym doors depend on it today. Computer Vision is the Phase 4 AI feature that watches a member's exercise form through a camera. Keeping these separate avoids a common mix-up.

4. Technology Stack

Layer	Technology	Why
Backend framework	Django + Django REST Framework	Mature, batteries-included, fast to build secure CRUD + APIs; strong ORM for complex relational data
Database	PostgreSQL	Relational integrity for tenant/member data; strong support for JSON fields (useful for flexible assessment data) and row-level isolation patterns
API style	REST, API-first	Every feature exposed via API from day one so web, mobile, and future AI services can all consume the same backend
Frontend (web)	React	Component reusability, large ecosystem, matches existing team experience
Mobile	Flutter	Single codebase for iOS + Android member-facing app
Background jobs / events	Celery + Redis (or equivalent)	Async processing for notifications, reminders, and the event bus

Layer	Technology	Why
Auth	JWT-based auth + RBAC middleware	Stateless auth suited to API-first + mobile; RBAC enforced at the API layer
Hosting	Cloud (AWS/GCP/Azure — TBD)	Scalable, supports later addition of GPU-backed services for CV/AI in Phase 3-4
AI / LLM (Phase 3)	AI Gateway + external LLM API(s)	Powers AI Coach and plan generation; gateway keeps the platform vendor-agnostic
Retrieval layer (Phase 3)	Vector DB (e.g. pgvector on Postgres)	Lets the AI Coach retrieve relevant Member 360 context before answering (RAG pattern)
Computer Vision (Phase 4)	Pose-estimation model (e.g. MediaPipe/OpenPose), edge-first inference	Real-time form feedback; processing kept close to the device for speed and privacy
Predictive models (Phase 4)	Lightweight ML (e.g. gradient boosting) on event history	Churn/retention scoring — doesn't need deep learning to be useful at this stage
Voice AI (Phase 4)	Telephony API + speech-to-text/text-to-speech + LLM	Powers the AI Sales Agent for calls; a distinct stack from the text-based AI Coach

5. Phase Roadmap

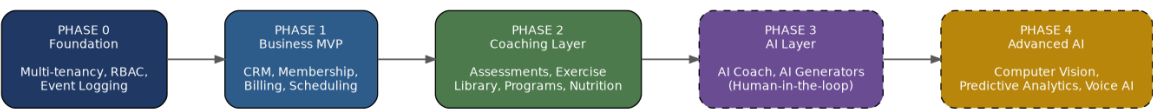


Figure 2 — Five-phase roadmap. Every phase below is detailed in full: diagram, modules, rationale, and exit criteria.

Phase	Goal	Depends on AI?
Phase 0 — Foundation	Multi-tenancy, RBAC, core data model, event logging	No
Phase 1 — Business MVP	CRM, membership, billing, scheduling — a sellable product	No
Phase 2 — Coaching Layer	Assessments, exercise library, manual program & diet builder	No
Phase 3 — AI Layer	AI Coach, AI-generated plans (trainer-approved), AI Gateway live	Yes — requires Phase 0-2 data
Phase 4 — Advanced AI	Computer vision form-check, churn prediction, voice AI sales agent	Yes — requires usage history from Phase 1-3

6. Phase 0 — Foundation

Purpose: establish the structural decisions that are extremely costly to change later — before any customer-facing feature is built.

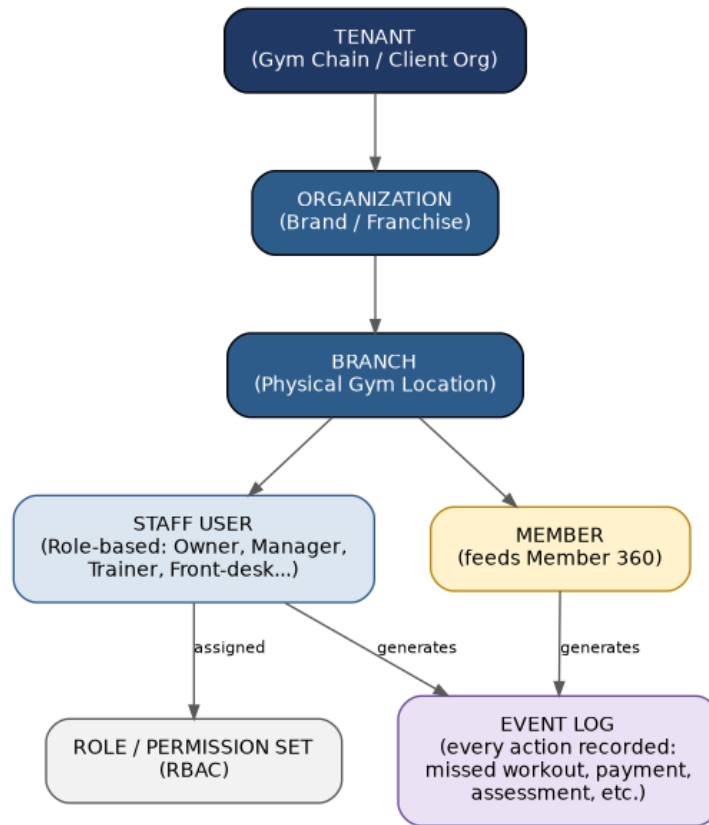


Figure 3 — Tenancy, RBAC and event-logging hierarchy established in Phase 0.

6.1 What gets built

- Tenant → Organization → Branch data hierarchy, with every downstream table scoped to a tenant
- Role-Based Access Control (RBAC): role and permission framework covering staff roles (Owner, Manager, Trainer, Front-desk, etc.)
- Core Member entity — the seed of the future Member 360 profile
- Event logging system: a central event bus/table that records meaningful actions across the platform
- Authentication (JWT-based), password policies, session handling
- Base API scaffolding and environment setup (dev/staging/prod)

6.2 Why this order matters

These four items — tenancy, RBAC, Member 360 seed, and event logging — are foundational because nearly every table and feature built afterward references them. Retrofitting tenant isolation or event logging into a system that already has hundreds of tables and business rules is one of the most expensive refactors in software; doing it first avoids that cost entirely.

6.3 Exit criteria for Phase 0

- A new tenant/organization/branch can be provisioned end-to-end

- Staff users can be created and assigned roles with enforced permissions
- A test event can be logged and queried against a member record
- Core API layer is live with authentication working

7. Phase 1 — Business Platform MVP

Purpose: deliver a complete, independently sellable gym/business management product — no AI required. This is the commercial foundation. Updated in this version to include Coupons, Referrals & Rewards, Facebook Lead Ads, and Access Control — all confirmed as real, active parts of the client's current business, not optional extras.

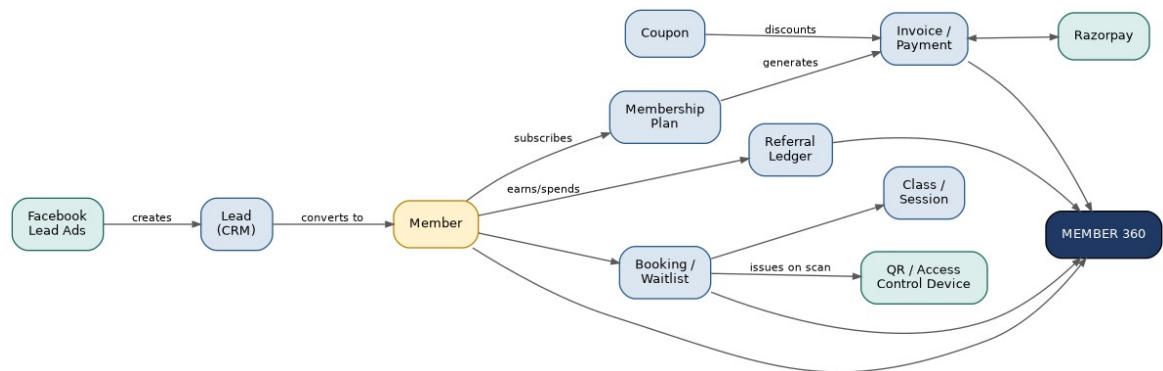


Figure 4 — Phase 1 module relationships, updated with coupons, referrals, Facebook lead ads, and access control.

7.1 What gets built

Module	Core capability
CRM & Lead Management	Capture leads, track follow-ups, convert leads to paying members. Facebook Lead Ads feeds in as a new lead source via the Integration Layer (Section 11)
Membership Management	Plans, pricing, renewals, freezes, upgrades/downgrades
Billing & Payments (incl. Coupons)	Invoicing, Razorpay integration, trainer commission tracking, and discount/coupon logic applied at checkout
Referrals & Rewards	New module. Tracks referral points/credit earned and spent — e.g. member refers a friend, gets a reward; built on the same event-logging pattern as the rest of the platform
Class & Schedule Management (incl. Access Control)	Class creation, booking, waitlists, trainer assignment. Includes the QR/card-based door entry flow: booking exists → QR generated → scanned at the door → attendance logged — the same pattern already proven in the client's current system
Staff / HR / Inventory (light)	Trainer attendance, basic stock tracking for the gym

7.2 Why these additions belong here, not later

Coupons are a natural extension of Billing, not a separate phase — a coupon has to be checked at the exact moment a price is calculated. Referrals & Rewards gets its own module because it has its own ledger and its own triggers (new referral, renewal, birthday), but it's still simple, rule-based logic — no AI needed. Facebook Lead Ads and Access Control are both external-service connections, not new business logic, so they're built as Integration Layer adapters (Section 11) feeding into CRM and Booking respectively — this keeps the core modules clean even as more external services get added later.

7.3 Why this comes before the AI/coaching features

A gym cannot operate without membership, billing, and scheduling — these are needed regardless of whether any AI feature ever ships. Building this first also means the platform has real, sellable value early, independent of how the AI phases progress.

7.4 Exit criteria for Phase 1

- A lead can be captured (manually or via Facebook Lead Ads), followed up, and converted into a paying member
- A member can be billed with a coupon applied, and a class can be booked, scanned via QR at the door, and attended
- A referral can be logged and correctly credited to the referring member
- All actions from this phase are visible in the Member 360 timeline

8. Phase 2 — Coaching Layer

Purpose: add the fitness/coaching content and workflows — still without AI. A human trainer builds and assigns everything; the system just structures and stores it well.

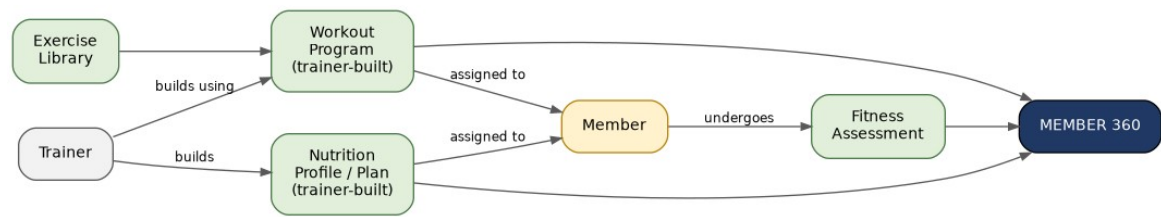


Figure 5 — Coaching layer module relationships. All AI in this diagram is deferred to Phase 3.

8.1 What gets built

Module	Core capability
Fitness Assessment	Structured intake: measurements, body composition, mobility, strength baselines
Exercise Library	Searchable database of exercises with instructions, media, difficulty, muscle groups
Workout Program Builder	Trainer assembles exercises into weekly/monthly programs, assigned to a member
Nutrition & Diet Management	Trainer-built meal plans and macro/calorie targets, including an Indian food dataset

8.2 Why this comes before AI generation

The AI Plan Generator (Phase 3) needs a well-structured Exercise Library and assessment data to generate anything useful — an AI cannot generate a good workout plan without a real data set to build from. Phase 2 is what makes Phase 3 possible: it's the content and workflow layer the AI will eventually sit on top of and accelerate.

8.3 Exit criteria for Phase 2

- A trainer can run a full assessment on a member and see it saved to Member 360
- A trainer can build a program from the Exercise Library and assign it to a member
- A trainer can build a diet plan and assign it to a member

9. Phase 3 — AI Layer

Purpose: switch on the first AI-powered features — an AI Coach and AI-generated workout/diet plans — built on top of the data and content created in Phase 0-2. This is where the AI Gateway becomes a real, running component rather than just an architectural placeholder.

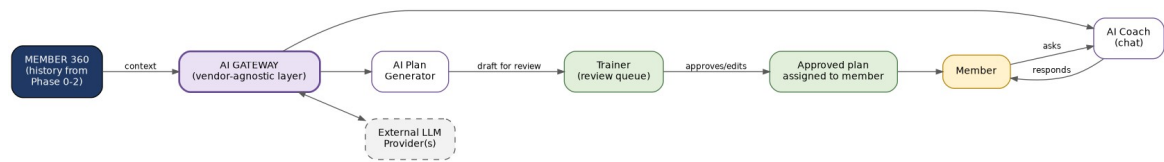


Figure 7 — Phase 3 flow: Member 360 context feeds the AI Gateway, which powers the AI Coach and Plan Generator, with a trainer approval step before anything reaches the member.

9.1 What gets built

Module	Core capability
AI Gateway	Central abstraction layer that routes requests to one or more LLM providers, so the rest of the platform never talks to a vendor API directly
AI Coach	Conversational assistant members can ask fitness/nutrition questions to, using their own Member 360 data as context
AI Plan Generator	Drafts a workout or diet plan using the Phase 2 Exercise Library and a member's assessment data
Trainer Review Queue	Every AI-generated plan lands here for a trainer to approve, edit, or reject before it reaches a member

9.2 Why this comes after Phase 0-2, not before

The AI Coach and Plan Generator are only as good as the data they draw on. Without a populated Exercise Library, structured assessments, and real Member 360 history from Phase 0-2, the AI has nothing meaningful to reason over — it would either produce generic, low-value output or need to be rebuilt once real data exists. Building the AI Gateway now, on top of a stable foundation, also means it's built once, correctly, rather than retrofitted.

9.3 Human-in-the-loop is a hard requirement, not a preference

No AI-generated plan reaches a member without a trainer approval step. This directly reflects the scope document's own three-tier model: AI can auto-execute low-risk actions (reminders, content suggestions), AI can recommend but a human must approve (workout/diet changes), and some decisions remain human-only (anything medical). Phase 3 architecture enforces this at the workflow level — the review queue is not optional and cannot be bypassed by configuration.

9.4 Exit criteria for Phase 3

- The AI Gateway can route a request to at least one LLM provider and return a response
- A member can ask the AI Coach a question and get a context-aware answer grounded in their own Member 360 data
- An AI-generated plan can be created, reviewed, edited, and approved by a trainer before assignment

- Switching the underlying LLM provider requires a configuration change, not a code change in the Coach or Generator modules

10. Phase 4 — Advanced AI

Purpose: add the most differentiated — and most technically demanding — capabilities: computer vision form-checking, predictive retention modeling, and a voice AI sales agent. These are sequenced last because each one depends on data, infrastructure, or trust built in earlier phases.

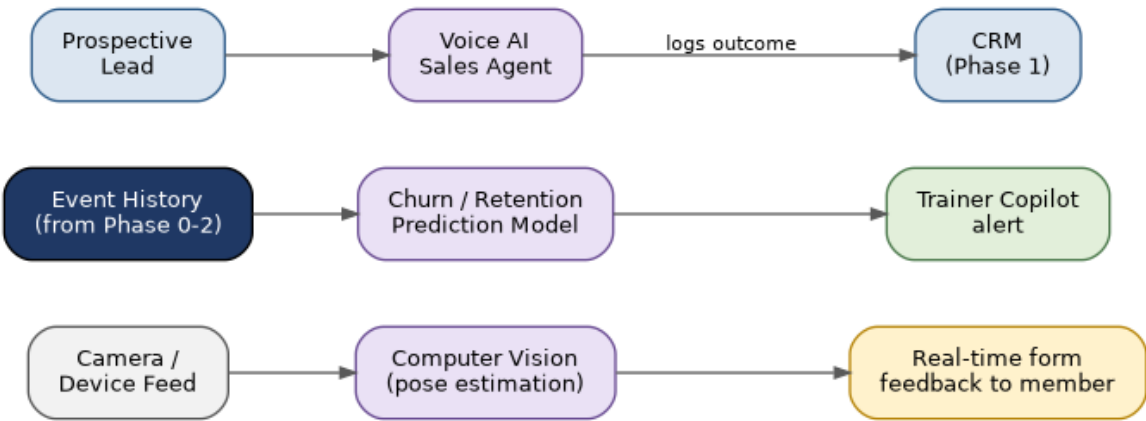


Figure 8 — Phase 4's three largely independent capabilities: computer vision, predictive analytics, and voice AI — each consuming data or infrastructure from earlier phases.

10.1 What gets built

Module	Core capability
Computer Vision (form analysis)	Uses a camera feed and pose-estimation to give a member real-time feedback on exercise form (e.g. “knees caving in”)
Predictive Analytics (churn/retention)	Scores members on likelihood to disengage, using event history accumulated since Phase 0, and alerts trainers via Copilot
Voice AI Sales Agent	Places or receives calls with prospective leads, qualifies them conversationally, and logs the outcome back into the Phase 1 CRM

10.2 Why this comes last

Each Phase 4 capability has a hard dependency that only exists after earlier phases are live:

- Computer vision needs the Exercise Library (Phase 2) to know what “correct form” looks like for a given movement
- Churn prediction needs months of real event history (Phase 0 onward) — a model trained on no data isn't useful, and this history cannot be created retroactively
- The voice AI sales agent needs the CRM (Phase 1) to log into, and benefits from the AI Gateway (Phase 3) already being a proven, stable pattern in production

10.3 Special considerations

- Computer vision inference is planned as edge-first (processed close to the member's device) rather than sent to the cloud, for both latency and privacy reasons
- Camera and biometric data require explicit, purpose-specific consent — this is a DPDP requirement, not just good practice (see Section 12.2)

- Voice AI calling should be scoped against telecom/consent regulations for outbound calls in each operating region before build

10.4 Exit criteria for Phase 4

- A member's exercise form can be analyzed live and specific, actionable feedback surfaced during the movement
- A churn-risk score is generated per member and surfaced to a trainer with enough lead time to act
- A voice AI call with a lead can be completed end-to-end, with the outcome correctly logged in the CRM

11. Integration Layer

Every external service the platform depends on — payments, messaging, calling, email, lead sources, and physical access control hardware — connects through one Integration Layer, using the adapter pattern: one clean interface per service, so core modules (Billing, CRM, Notifications, Booking) never talk to a vendor's API directly.

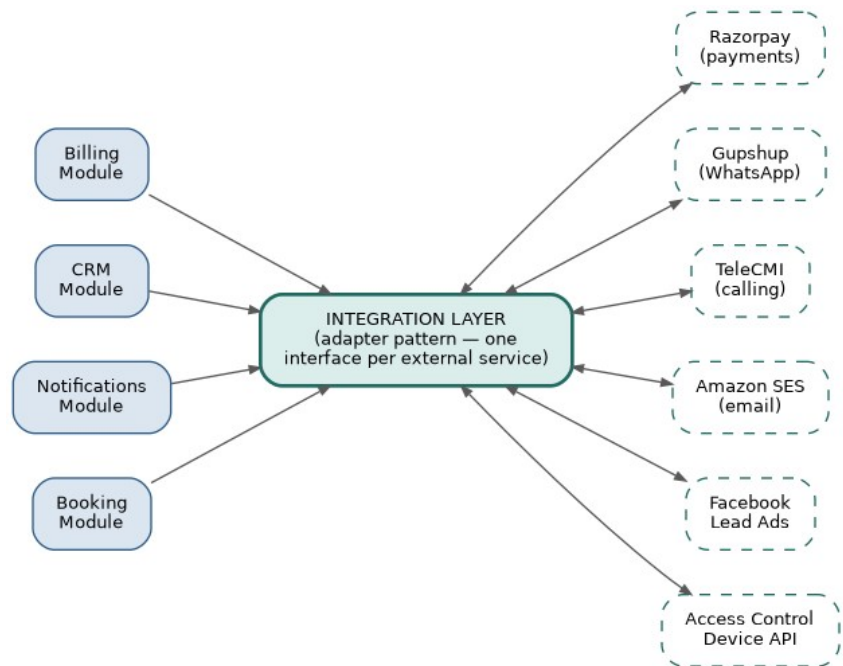


Figure 9 — The Integration Layer sits between core modules and every external service, so swapping or adding a vendor later is a config change, not a rewrite.

11.1 Adapters needed (confirmed from client discovery)

Adapter	Used by	Purpose
Razorpay	Billing	Payment processing — already in use by the client, reuse the same merchant account
Gupshup	Notifications, CRM	WhatsApp messaging — booking confirmations, reminders, follow-ups
TeleCMI	CRM	Calling/telephony for lead follow-up; also the foundation for the Phase 4 AI voice agent
Amazon SES	Notifications	Transactional email
Facebook Lead Ads	CRM	Inbound lead capture from Facebook lead forms, feeding directly into the Leads module
Access Control Device API	Booking/Scheduling	Door entry hardware — create temporary user/card on booking, delete after use, exactly matching the client's proven current flow

11.2 Why this is one layer, not scattered integrations

The client's current system hardcodes some of these connections directly into business logic (for example, the device API key sits inside a booking service file, not a config value). This makes future changes — a new payment provider, a different messaging vendor — expensive and risky. Centralizing all of this now means Phase 1 development is barely harder, but every future integration change becomes isolated and low-risk.

11.3 Access Control adapter — specific design notes from the client's proven flow

The client's existing system already validated a specific, working pattern for door access, worth replicating exactly rather than redesigning:

- A booking must exist and be valid before a QR/card can be generated — access control never bypasses booking rules
- QR codes are time-boxed around the class start/end time, using configurable timing values — not hardcoded
- On successful scan, the door device is updated via API, and the temporary user/card is deleted afterward so access can't be reused outside the booked window
- The device-to-class mapping is per-plan (multiple doors/devices possible), not a single global device

12. Existing System & Migration Plan — Sweatfit Wellness

This section is specific to onboarding the first real client, Sweatfit Wellness, whose current system (“BPC Admin”) needs to be understood and migrated without losing data. This is additive to the phase plan above — it doesn't change Phase 0-4, it defines how Tenant #1 gets onto the platform.

12.1 What the existing system actually is

Not one custom system, but several pieces: a WordPress marketing site, a small custom React checkout app, a Node.js/Express + MySQL backend (“BPC Admin”) handling bookings/CRM/payments, and a separate Windows-based access control service talking to physical door hardware. The core business logic — users, plans, bookings, transactions — lives in BPC Admin's MySQL database, confirmed via a 65-table schema export.



Figure 10 — Migration approach: export and clean, run in parallel to verify, then cut over — never a single risky switch.

12.2 Data migration mapping (initial draft — pending source code/API confirmation)

Existing table (BPC Admin)	New platform equivalent	Note
users	Member (Member 360 core)	Status is currently spread across multiple flags (istrial, isverified, block) — decide whether to consolidate into one status field
plans, sessions, types	Membership Plan catalog	Fairly direct mapping
transactions	Invoice / Payment	transactionid (string) and uid (internal key) are different — don't conflate during migration
userplans	Active membership entitlement	No single status column today — status is calculated from dates and sessions remaining; consider adding a real status field in the new schema
class, slots, bookings	Class & Schedule module	Preserve the public booking reference (book_id) separately from the internal numeric key, same as the existing system
coupons2	Coupons (Billing extension)	Use the newer coupons2 structure as the reference — coupons (v1) and coupons_history are legacy
referral_transactions	Referrals & Rewards ledger	Direct mapping — confirm exact trigger events (birthday, renewal, referral) with the client
leads, LeadDetailsView	CRM Leads module	Direct mapping
configurations	Platform rules/config table	Bring across QR timing values and booking limits explicitly — these are tuned, live business rules
external_device	Access Control adapter	Confirm current device host/IP list matches the

Existing table (BPC Admin)	New platform equivalent	Note
	config	access-control handover PDFs
users2, archive_users, userplans2, userplans_backup, userplans_receipt, coupons, pilates_tempdb, and similar	Not migrated by default	Legacy/duplicate tables — confirmed still present in the schema export; each needs manual confirmation before inclusion

12.3 What migrates as-is vs. what gets redesigned

- Migrate as-is: the access control flow (booking → QR → scan → webhook → delete), Razorpay account/merchant setup, Gupshup/SES/TeleCMI vendor accounts
- Redesign: the core booking/member/CRM data model — rebuilt cleanly inside this platform's schema rather than copying the existing structure, since the existing schema has very few enforced database-level relationships and several legacy duplicate tables
- Rebuild, don't carry over: the small join.sweatfitwellness.com checkout app — folded into this platform's own Billing/checkout flow instead of staying a separate app

12.4 Migration sequencing

1. Get read access to source code, API docs, and a sanitized (non-production) data export
2. Migrate historical/static data first — payments, past bookings — into the new platform, verify against the old system
3. Run both systems in parallel — new platform live for staff use, old system still handling real member traffic
4. Cut over booking/CRM traffic once data is verified matching
5. Cut over the access control webhook last, since it's hardware-dependent and affects real people at the door — repoint the existing device flow's webhook URL to the new Access Control adapter, keeping the exact same create → scan → webhook → delete pattern
6. Decommission the old system only after a full billing cycle has run cleanly on the new one

13. Why Event Logging Starts Now: A Worked Example

This is a concrete illustration of why Phase 0's event logging matters even though no AI exists yet in Phase 0-2.



Figure 6 — An event logged in Phase 0-2 becomes usable AI input in Phase 3-4, but only if it was captured at the time.

If “workout_missed” events aren't logged from day one, there is no historical record for a future churn-prediction model to learn from — that history cannot be recreated retroactively. Every month of Phase 1-2 operation without event logging is a month of lost training data for Phase 3-4.

14. Cross-Cutting Concerns

14.1 Security (built into every phase)

- TLS in transit, encryption at rest for sensitive fields
- RBAC enforced at the API layer, not just the UI
- Tenant isolation enforced at the query layer (not just application logic)
- Full audit logging of access to sensitive member data

14.2 DPDP Act, 2023 — Status

The architecture supports the core DPDP concepts — consent capture, data minimization, retention rules, and data export/deletion — as part of the Member 360 and event design. The following statutory items are not yet addressed and should be flagged as follow-up items, ideally with legal input, before public launch:

- Breach notification process (to the Data Protection Board and affected users)
- Cross-border data transfer rules (relevant once any AI vendor, e.g. a US-based LLM API, or cloud host is chosen — relevant from Phase 3 onward)
- Grievance Officer / DPO designation for data-rights complaints specifically (distinct from general customer support)
- Parental consent handling for members under 18
- Explicit, purpose-specific consent for camera/biometric data ahead of Phase 4 computer vision

Confirmed, not hypothetical: sensitive health data already exists

- The client's existing system (Section 12) includes PAR-Q intake forms storing medical conditions, medications, surgeries, and pregnancy/postpartum status — real, live data, not a future concern.
- This data must be migrated (if at all) with explicit care — encrypted, access-restricted, and covered by clear consent language before it moves into the new platform.
- Treat this as a required legal review item before Phase 1 migration of member data begins, not something to revisit later.

14.3 SOC 2 — Status

SOC 2 is not a certification a document or architecture plan can grant — it requires an independent auditor to verify security controls operate correctly over an extended period (often 3-12 months of live evidence). This architecture is

designed to make a future SOC 2 audit achievable (RBAC, encryption, tenant isolation, audit logs are all in scope from Phase 0), but formal SOC 2 pursuit should be treated as a later milestone, typically triggered by enterprise/B2B customer requirements rather than a phase-specific deliverable.

14.4 AI Gateway — Design Now, Activate in Phase 3

The AI Gateway is fully designed as part of this plan (Section 9.1) but only becomes an active, running component in Phase 3. The module boundaries in Phase 1-2 are deliberately kept clean specifically so the Gateway can be introduced without reworking earlier modules — this is why it's architected alongside Phase 0-2 even though it isn't switched on until later.

15. Phase Sequencing Rationale

Every capability in the original 60-section scope is represented somewhere in Phase 0-4 above — nothing has been dropped. What this plan does is impose a build order based on real dependencies, so the team isn't asked to build AI features before there's data for them to use, or computer vision before there's an Exercise Library to validate form against. The table below summarizes why each AI/advanced capability sits where it does.

Capability	Phase	Hard dependency
AI Coach / chatbot	3	Member 360 history (Phase 0-2) to give it something to reason over
AI-generated workout/diet plans	3	Exercise Library + assessments (Phase 2) as source content
Computer vision form analysis	4	Exercise Library (Phase 2) defines what “correct form” means
Churn / retention prediction	4	Months of real event history (Phase 0 onward) to train against
AI Sales Agent / voice AI	4	CRM (Phase 1) to log into; proven AI Gateway pattern (Phase 3)
AI Business Intelligence (NL queries)	4	Sufficient operational data across all prior phases to query meaningfully

16. Risks & Dependencies

Risk	Mitigation
Multi-tenancy design flaws surface late	Dedicated schema review before any Phase 1 module starts; automated tests for tenant data isolation
Scope creep — AI/CV work starting during Phase 1-2	Explicit phase gate reviews; Phase 3-4 work not started until Phase 0-2 exit criteria are met
DPDP gaps discovered post-launch	Legal review of data flows scheduled before public launch, and specifically before member health data (PAR-Q) or Phase 4 camera/voice features are touched
Exercise Library / assessment data not structured well enough for future AI use	Design data model in Phase 2 with Phase 3 AI consumption in mind, reviewed against Section 2.2 principle
LLM/AI provider cost or reliability issues (Phase 3)	AI Gateway design allows switching or load-balancing across providers without touching application code
Computer vision accuracy/liability concerns (Phase 4)	Position CV feedback as coaching guidance, not medical/injury diagnosis; human trainer remains the authority
Voice AI / outbound calling compliance (Phase 4)	Review telecom consent and do-not-call regulations per operating region before build, not after

Risk	Mitigation
Client data ownership/export rights unclear (BPC Admin system)	Confirm in writing who can grant DB access/export before estimating migration timeline
Legacy/duplicate tables migrated by mistake (users2, userplans_backup, etc.)	Explicit exclusion list (Section 12.2); manual confirmation required before any legacy table is included
Very few DB-level foreign keys in existing system — relationships only enforced in application code	Migration scripts must validate relationships explicitly; do not assume referential integrity from the old system

17. Open Questions — Confirm Before Building

Running list of things that are not yet known and should not be assumed. Update this section as answers come in rather than guessing ahead of confirmation.

17.1 About the existing client system

- Who has direct database/server access to BPC Admin — the client, an agency, or a vendor?
- Is app.sweatfitwellness.com confirmed to be running on the BPC Admin backend?
- Current, correct webhook URLs — two slightly different domains appeared across documents (admin-nodelive.bodyprocoach.com vs adminapi.bodyprocoach.com)
- Why does the hardcoded “card 1001 / guest” exception exist, and should it carry into the new system?
- Which legacy/duplicate tables are genuinely dead vs. quietly still in use?
- Exact referral/reward trigger rules (does referral credit expire? how are agent commissions split?)

17.2 Still needed from the client/senior dev

- Source code repository access
- API documentation or a Postman collection
- Environment variable / credentials list (to be rotated before reuse, not copied as-is)
- A sanitized, non-production data export to build and test migration scripts against

17.3 About scope

- Should Facebook Lead Ads and Access Control be treated as reusable, sellable modules for future clients, or built Sweatfit-specific first and generalized later?
- Does the referral/commission structure vary enough across future clients to need per-tenant configuration from day one?

18. Next Steps for Development

- Start Phase 0 (Section 6): tenancy, RBAC, Member 360 seed, event logging, base API scaffolding
- Stand up the Integration Layer skeleton (Section 11) early, even before every adapter has real credentials — the interface pattern matters more than having all six wired up immediately
- Design the Phase 1 schema using Section 12.2's mapping table as a starting reference, not a copy of the existing structure
- Escalate Section 17's open questions to senior dev/client before starting migration-specific code
- Treat this document as living — update Section 17 as answers come in, and revisit Section 12 once source code/API docs are available